

# Certifiable Trajectory Optimization for Rigid-Body Systems on Lie Groups via Semidefinite Relaxation

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Final Report Master's Thesis

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Harvard University



[Sankaran 2022]



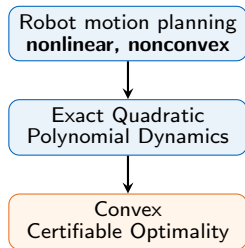
[Thermal Processing 2024]

**Robotic systems operating in complex environments require motion planning methods that are both dynamically accurate and computationally reliable.**

# Motivation

How can robots plan reliable motions in complex environments?

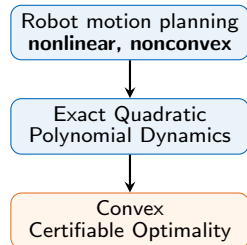
- Nonlinear dynamics and geometric constraints  
⇒ nonconvex motion planning problems [Kang+ 2024].
- Classic global optimization methods simplify dynamics  
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How can robots plan reliable motions in complex environments?

- Nonlinear dynamics and geometric constraints  
⇒ nonconvex motion planning problems [Kang+ 2024].
- Classic global optimization methods simplify dynamics  
⇒ lower model fidelity [Teng+ 2025].
- Variational discretization **preserves** rigid-body dynamics.
- Semidefinite relaxation enables **convex** optimization.



## Key idea

Obtaining globally optimal solutions using Lie group variational discretization and semidefinite relaxation.

# Related Work

Lie Groups [Hall 2000]



Discrete Mechanics and Variational Integrators [Marsden+ 2001; Lee 2008]

Maximal Coordinates [Brüdigam+ 2021]

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## Research Question

How reliably can LGVI-SDP produce feasible and certifiably trajectories?

# Lie Group Variational Integrators

## Continuous Mechanics:

$$L(g, \xi) = T(g, \xi) - U(g)$$

(Lagrangian)

$g, f \in G$ : config.

$\xi \in \mathfrak{g}$ : velocity

## Discrete Mechanics: [Marsden+ 2001]

$$L_d(g_k, g_{k+1}) \approx \int_{t_k}^{t_{k+1}} L(g, \xi) dt$$

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$$D_1 L_d(g_k, g_{k+1}) + D_2 L_d(g_{k-1}, g_k) + h u_k = 0 \quad g_{k+1} = g_k f_k \quad (\text{Euler-Lagrange})$$



Preserves mechanical structure and evolves on the Lie group

$g, f \in G$ : config.

$\xi \in \mathfrak{g}$ : velocity

$u \in \mathfrak{g}^*$ : control force

$\eta \in \mathfrak{g}$ : variation

$D$ : derivative

$h \in \mathbb{R}$ : step size

# Multibody Systems in Maximal Coordinates

- Minimal coordinates: independent joint coordinates
- Maximal coordinates: position and orientation of every rigid body

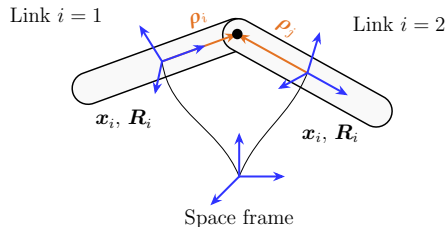
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**Configuration:** [Brüdigam+ 2021]

Body:  $q_i = (\mathbf{x}_i, \mathbf{R}_i) \in \mathbb{R}^d \times SO(d)$

Joint:  $\phi_{ij}(q) = \mathbf{x}_i + \mathbf{R}_i \boldsymbol{\rho}_i - \mathbf{x}_j - \mathbf{R}_j \boldsymbol{\rho}_j = 0$



Adapted from [Brüdigam+ 2021].

$q$ : full configuration

$\boldsymbol{\rho}_i$ : body-fixed joint vector

$i, j$ : rigid-body indices

$d$ : spatial dimension

$\mathbf{x}_i$ : position

$n_b$ : number of rigid bodies

$\mathbf{R}_i$ : orientation

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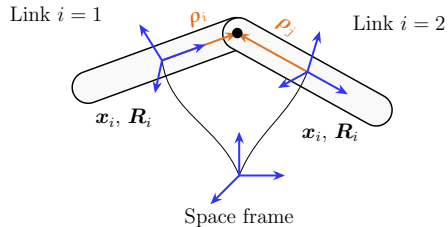
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**Discrete Action Sum:** [Brüdigam+ 2021]

$$S_d^{\max} = S_d + \frac{h}{2} \sum_{k=0}^{N-1} (\boldsymbol{\lambda}_k^\top \phi(q_k) + \boldsymbol{\lambda}_{k+1}^\top \phi(q_{k+1}))$$

$\boldsymbol{\lambda}_k$ : Lagrange multiplier



Adapted from [Brüdigam+ 2021].

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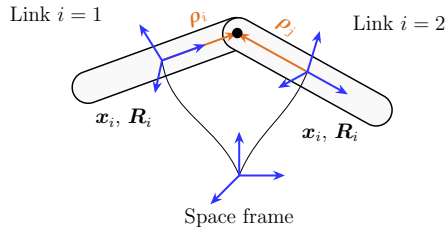
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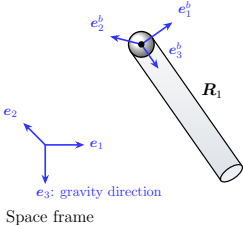


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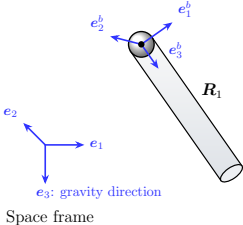
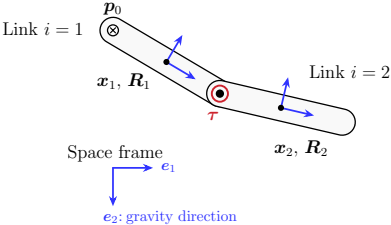
## Result

Maximal coordinates keep the discrete multibody dynamics quadratic.

# Rigid-Body Systems

|           | 3D-Pendulum                                                                                          | Acrobot |
|-----------|------------------------------------------------------------------------------------------------------|---------|
| System    |  <p>Space frame</p> |         |
| Variables | $(R_k, F_k, \bar{u}_{p,k}) \in SO(3)^2 \times \mathbb{R}^3$                                          |         |
| Size      | $21N + 6$                                                                                            |         |

# Rigid-Body Systems

|           | 3D-Pendulum                                                                                                                                     | Acrobot                                                                                                                                                                                                        |
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| System    |  <p>Space frame</p> <p><math>e_3</math>: gravity direction</p> |  <p>Link <math>i = 1</math></p> <p>Link <math>i = 2</math></p> <p>Space frame</p> <p><math>e_2</math>: gravity direction</p> |
| Variables | $(R_k, F_k, \bar{u}_{p,k}) \in SO(3)^2 \times \mathbb{R}^3$                                                                                     | $(q_{i,k}, F_{i,k}, \bar{\lambda}_{0,k}, \bar{\lambda}_{12,k}, \bar{u}_k) \in (\mathbb{R}^2 \times SO(2))^2 \times SO(2)^2 \times \mathbb{R}^5$                                                                |
| Size      | $21N + 6$                                                                                                                                       | $17N + 3 \rightarrow 13N - 1$<br>$x_{i,k}$ expressed through $R_{i,k}$ using joint constraints $\phi$                                                                                                          |

# Rigid-Body Systems

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## Result

The models preserve the geometric structure, stay on the Lie group, and remain quadratic.

# Optimization via Sparse Semidefinite Relaxation

LGVI dynamics  
on Lie groups

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Polynomial  
Optimization

## Polynomial Optimization

General problem: [Teng+ 2025]

$$p^* = \min_{z \in \mathbb{R}^{n_z}} p(z)$$

$$\text{s.t. } h_\ell(z) = 0$$

$$g_j(z) \geq 0$$

Nonconvex

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## Sparse SDP Relaxation

Relaxed problem: [Kang+ 2025]

$$p_{\text{SDP}}^* = \min_{X \in \mathbb{S}^n} \langle C, X \rangle$$

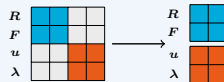
$$\text{s.t. } \mathcal{A}(X) = b, \quad X \succeq 0$$

Dense relaxation:

$$M_\kappa \succeq 0$$

Sparse relaxation:

$$M_{\kappa,1} \succeq 0, \dots, M_{\kappa,n_c} \succeq 0$$



Convex SDP

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SDP lower bound  
Trajectory

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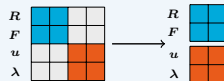
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Convex SDP

## Certification [Teng+ 2025]

Bounds:

$$J_{\text{SDP}} \leq p^* \leq J_{\text{NLP}}$$

Gap:

$$\epsilon = \frac{|J_{\text{NLP}} - J_{\text{SDP}}|}{|J_{\text{NLP}}| + 10^{-6}}$$

Rank:

$$\delta_{\max} = \max_k \frac{\nu_{2,k}}{\nu_{1,k}}$$

Certificate:

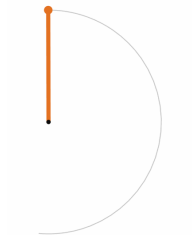
$$\epsilon \approx 0 \quad \text{and} \quad \delta_{\max} \approx 0$$



Globally optimal trajectory

# 3D-Pendulum Optimization Results

Parameters:  $R^{\text{init}} = R_x(5^\circ)$ ,  $F^{\text{init}} = I$ ,  $N = 20$ ,  $u_{\text{max}} = 5 \text{ N m}$



Trajectory  $R_{\text{des}} = R_x(180^\circ)$

| Metrics                                         | 180°                 |
|-------------------------------------------------|----------------------|
| Gap $\epsilon$                                  | $2.8 \times 10^{-9}$ |
| Rank $\delta_{\text{max}}$                      | $2.3 \times 10^{-8}$ |
| Manifold error $e_{\text{orth}}^{P,\text{max}}$ | $4.5 \times 10^{-8}$ |
| Runtime                                         | 64.6 min             |

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Trajectory  $R_{\text{des}} = R_x(180^\circ)$

| Target    | 0° | 45° | 90° | 135° | 180° |
|-----------|----|-----|-----|------|------|
| Certified | ✓  | ✓   | ✓   | ✓    | ✓    |

## Result

Global optimality is certified for all evaluated discrete 3D-pendulum POPs.

# Acrobot Optimization Results

**Parameters:**  $R_i^{\text{init}} = R(5^\circ)$ ,  $F_i^{\text{init}} = I$ ,  $i \in \{1, 2\}$ ,  $N = 60$ ,  $\Delta t_{\text{opt}} = 0.1 \text{ s}$ ,  $u_{\text{max}} = 15 \text{ N m}$

- **Chain sparsity:** time-local variable clique containing the coupled Acrobot dynamics and kinematics

$$M_{\kappa,k} \in \mathbb{S}^{253}$$

- **Multibody sparsity:** time-local clique is separated into body dynamics and joint kinematics

$$M_{\kappa,k}^{\text{dyn}} \in \mathbb{S}^{171}, \quad M_{\kappa,k}^{\text{kin}} \in \mathbb{S}^{91}$$

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|               | Chain sparsity |      | Multibody sparsity |      |
|---------------|----------------|------|--------------------|------|
| Target        | 45°            | 180° | 45°                | 180° |
| Certified     | ✓              | ✗    | ✓                  | ✗    |
| Runtime [min] | 59.7           | 90.4 | 17.6               | 31.2 |

- I. Chain  $\longrightarrow$  multibody sparsity:  $\approx 3\times$  **faster**
- II.  $\delta_1 < \delta_{\text{max}}$ : the first-step clique is **tighter** than the worst trajectory clique

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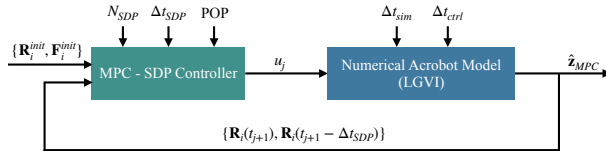
- Chain  $\longrightarrow$  multibody sparsity:  $\approx 3\times$  **faster**
- $\delta_1 < \delta_{\text{max}}$ : the first-step clique is **tighter** than the worst trajectory clique

## Result

Only cases  $\leq 45^\circ$  give feasible SDP extractions and globally certified trajectories

# Closed-Loop Acrobot Optimization Results

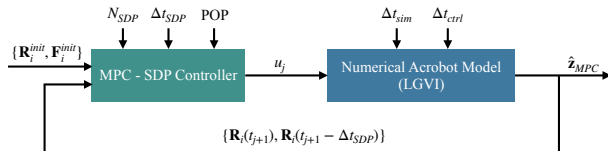
**Goal:** stabilize links at inverted equilibrium:  $\theta_1^{\text{des}} = \theta_2^{\text{des}} = 180^\circ$ ,  $\mathbf{F}_1^{\text{des}} = \mathbf{F}_2^{\text{des}} = \mathbf{I}$ ,  $u_{\text{max}} = 15 \text{ N m}$



SDP-based receding-horizon control loop.

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## Time-scale cases

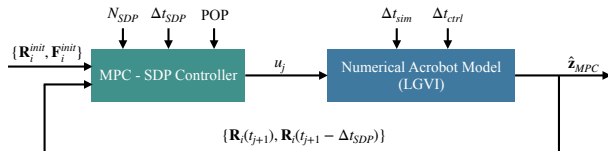
| Case | $\Delta t_{\text{SDP}}$ | $\Delta t_{\text{sim}}$ | $\Delta t_{\text{ctrl}}$ | $N_{\text{SDP}}$ |
|------|-------------------------|-------------------------|--------------------------|------------------|
| I    | 0.1                     | 0.1                     | 0.1                      | 20               |
| II   | 0.05                    | 0.05                    | 0.05                     | 40               |
| III  | 0.1                     | 0.0025                  | 0.025                    | 20               |

All time intervals in seconds.

Cases I-II: matched Case III: mismatched

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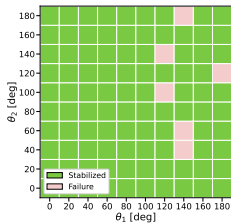
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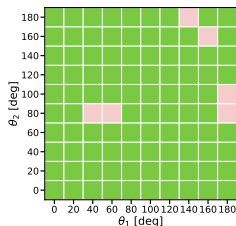
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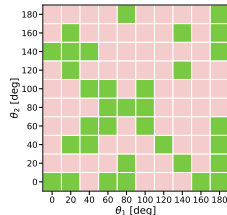
## Sampled initial conditions



Case I: 94/100 stabilized



Case II: 94/100 stabilized



Case III: 34/100 stabilized

# Closed-Loop Acrobot Optimization Results

**Goal:** Stabilization

**Initial condition:**  $\theta_1^{\text{init}} = 60^\circ$ ,  $\theta_2^{\text{init}} = 100^\circ$

| Case | $\Delta t_{\text{SDP}}$ | $\Delta t_{\text{sim}}$ | $\Delta t_{\text{ctrl}}$ | Time scales | $t_{\text{stab}}$ | Runtime |
|------|-------------------------|-------------------------|--------------------------|-------------|-------------------|---------|
| I    | 0.1 s                   | 0.1 s                   | 0.1 s                    | matched     | 2.3 s             | 2.7 h   |
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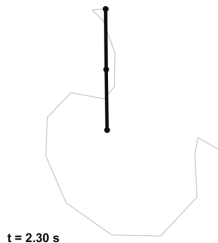
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| I    | 0.1 s                   | 0.1 s                   | 0.1 s                    | matched     | 2.3 s             | 2.7 h   |
| III  | 0.1 s                   | 0.0025 s                | 0.025 s                  | mismatched  | 6.775 s           | 19.7 h  |

**Case I: matched time scales**



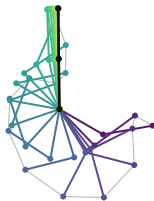
# Closed-Loop Acrobot Optimization Results

**Goal:** Stabilization

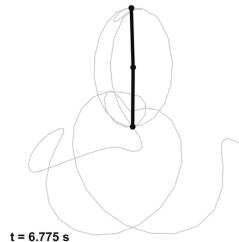
**Initial condition:**  $\theta_1^{\text{init}} = 60^\circ$ ,  $\theta_2^{\text{init}} = 100^\circ$

| Case | $\Delta t_{\text{SDP}}$ | $\Delta t_{\text{sim}}$ | $\Delta t_{\text{ctrl}}$ | Time scales | $t_{\text{stab}}$ | Runtime |
|------|-------------------------|-------------------------|--------------------------|-------------|-------------------|---------|
| I    | 0.1 s                   | 0.1 s                   | 0.1 s                    | matched     | 2.3 s             | 2.7 h   |
| III  | 0.1 s                   | 0.0025 s                | 0.025 s                  | mismatched  | 6.775 s           | 19.7 h  |

**Case I: matched time scales**



**Case III: mismatched time scales**



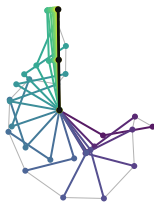
# Closed-Loop Acrobot Optimization Results

**Goal:** Stabilization

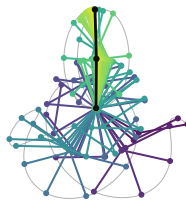
**Initial condition:**  $\theta_1^{\text{init}} = 60^\circ$ ,  $\theta_2^{\text{init}} = 100^\circ$

| Case | $\Delta t_{\text{SDP}}$ | $\Delta t_{\text{sim}}$ | $\Delta t_{\text{ctrl}}$ | Time scales | $t_{\text{stab}}$ | Runtime |
|------|-------------------------|-------------------------|--------------------------|-------------|-------------------|---------|
| I    | 0.1 s                   | 0.1 s                   | 0.1 s                    | matched     | 2.3 s             | 2.7 h   |
| III  | 0.1 s                   | 0.0025 s                | 0.025 s                  | mismatched  | 6.775 s           | 19.7 h  |

**Case I: matched time scales**



**Case III: mismatched time scales**



## Result

Dynamically feasible stabilization trajectories, with an high success rate for matched time scales

## Limitation

No global optimality certificate is obtained, and mismatched time scales reduce stabilization

# Conclusion and Outlook

## Conclusion

- Derived LGVIs for the 3D pendulum and Acrobot
- Relaxed nonconvex POPs into sparse convex SDPs
- Evaluated the framework in open-loop and closed-loop

## Outlook

- Analyze unit quaternions for smaller  $SO(3)$  moment blocks
- Evaluate GPU-accelerated first-order SDP solvers (e.g., CuADMM)
- Extend to larger systems, obstacles, contacts, and hardware experiments

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# Appendix

## Single-Body SDP Optimization

- All five 3D-pendulum targets generate feasible, **globally certified** trajectories.
- The  $180^\circ$  case requires 64.6 min. The method is not real-time capable.

**Certified, but offline**

## Multibody SDP Optimization

- **Only small rotations** ( $\leq 45^\circ$ ) are feasible, globally certified trajectories.
- Multibody cliques reduce runtime by 3-5 $\times$ , but do not guarantee tightness.

**Certification problem-dependent**

## Closed-Loop Multibody SDP Optimization

- First-control extraction generates **feasible swing-ups** despite non-tight predictions.
- Matched time scales stabilize 94/100 states; mismatch reduces this to 34/100.
- Decreasing time-matched  $\Delta t_{\text{SDP}} = \Delta t_{\text{sim}}$  requires larger  $N_{\text{SDP}}$ : **computational expensive**.

**Feasible, uncertified, and offline**

# Lie Groups and Lie Algebras

## Lie groups for rigid-body motion

$$SO(n) = \{ \mathbf{R} \in \mathbb{R}^{n \times n} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}, \det(\mathbf{R}) = 1 \}$$

$$SE(n) = \left\{ \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \mid \mathbf{R} \in SO(n), \mathbf{p} \in \mathbb{R}^n \right\}$$

## Lie algebras: local tangent representation

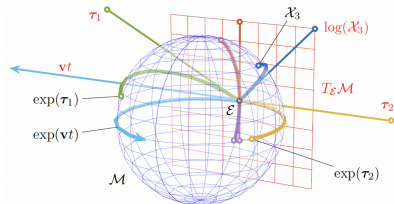
$$\mathfrak{so}(n) = \{ \boldsymbol{\omega} \in \mathbb{R}^{n \times n} \mid \boldsymbol{\omega}^\top = -\boldsymbol{\omega} \}$$

$$\mathfrak{se}(n) = \left\{ \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\omega} & \mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix} \mid \boldsymbol{\omega} \in \mathfrak{so}(n), \mathbf{v} \in \mathbb{R}^n \right\}$$

$$\exp : \mathfrak{so}(n) \rightarrow SO(n), \quad \exp : \mathfrak{se}(n) \rightarrow SE(n)$$

- Lie groups describe nonlinear configuration spaces.
- Lie algebras provide local coordinates for velocities and variations.

## Manifold viewpoint



### Variations on Lie Groups

$$\mathbf{R}(\epsilon) = \mathbf{R} \exp(\epsilon \hat{\boldsymbol{\eta}})$$

$$\delta \mathbf{R} = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathbf{R}(\epsilon) = \mathbf{R} \hat{\boldsymbol{\eta}}$$

Variations are taken in the tangent space and mapped back to the group.

# 3D-Pendulum on $SO(3)$

**Lagrangian Discretization:** [Lee 2008]

$$L(\mathbf{R}, \hat{\boldsymbol{\omega}}) = \underbrace{\frac{1}{2} \text{tr}[\hat{\boldsymbol{\omega}} \mathbf{J}_d \hat{\boldsymbol{\omega}}^\top]}_{T(\mathbf{R}, \hat{\boldsymbol{\omega}})} + \underbrace{m g \mathbf{e}_3^\top \mathbf{R} \boldsymbol{\rho}_c}_{U(\mathbf{R})}$$

angular velocity approx.  $\Downarrow$  trapezoidal approx.

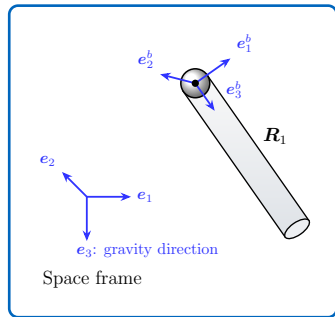
$$L_d(\mathbf{R}_k, \mathbf{F}_k) = \frac{1}{h} \text{tr}[(\mathbf{I} - \mathbf{F}_k) \mathbf{J}_d] - \frac{h}{2} (U(\mathbf{R}_k) + U(\mathbf{R}_k \mathbf{F}_k))$$

**Discrete Euler-Lagrange:**

$$\text{(Dynamics)} \quad \frac{1}{h} \left( \mathbf{F}_{k+1} \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_{k+1}^\top - \mathbf{J}_d \mathbf{F}_k + \mathbf{F}_k^\top \mathbf{J}_d \right) = h (\mathbf{M}_{k+1} + \mathbf{u}_{k+1})^\wedge$$

$$\text{(Kinematics)} \quad \mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k$$

**Configuration Manifold:**



# Acrobot on $(\mathbb{R}^2 \times SO(2))^2$

## Discrete Lagrangian in Maximal Coordinates:

$$L_d(\mathbf{q}_k, \mathbf{q}_{k+1}) = T_{d,T} + T_{d,R} - V_{d,T}$$

## Constrained Discrete Euler–Lagrange Equations:

$$D_2 L_d(\mathbf{q}_{k-1}, \mathbf{q}_k) + D_1 L_d(\mathbf{q}_k, \mathbf{q}_{k+1}) + h G(\mathbf{q}_k)^\top \lambda_k + \mathbf{f}_{d,k}(\tau_k) = \mathbf{0}$$

$\delta \mathbf{x}_{i,k} \implies$  translational dynamics

$\delta \mathbf{R}_{i,k} \implies$  rotational dynamics

Holonomic joints:  $\phi_{0,k} = 0, \phi_{12,k} = 0$

Kinematics:  $\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}$

$$\mathbf{G}(\mathbf{q}_k) = D\phi(\mathbf{q}_k), \quad \tau_k = [-u_k \ u_k]^\top$$

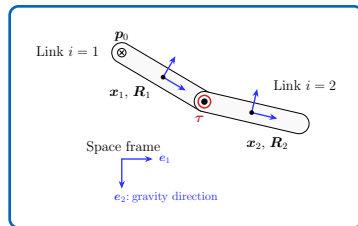
## Reduction Using the Holonomic Constraints:

$$\mathbf{x}_{1,k} = \mathbf{p}_0 - \mathbf{R}_{1,k} \boldsymbol{\rho}_{1,0}, \quad \mathbf{x}_{2,k} = \mathbf{p}_0 + \mathbf{R}_{1,k}(\boldsymbol{\rho}_{1,12} - \boldsymbol{\rho}_{1,0}) - \mathbf{R}_{2,k} \boldsymbol{\rho}_{2,12}$$

$$\mathbf{x}_{i,k} \text{ eliminated} \implies 17N + 3 \rightarrow 13N - 1 \text{ variables}$$

dynamics remain polynomial.

## Configuration Manifold:



$$\mathbf{q}_k = (\mathbf{x}_{1,k}, \mathbf{R}_{1,k}, \mathbf{x}_{2,k}, \mathbf{R}_{2,k})$$

# Polynomial Rotation Constraints

## Planar Rotation Variables

$$\mathbf{R}_{i,k} = \begin{bmatrix} c_{i,k} & -s_{i,k} \\ s_{i,k} & c_{i,k} \end{bmatrix}, \quad \mathbf{F}_{i,k} = \begin{bmatrix} a_{i,k} & -b_{i,k} \\ b_{i,k} & a_{i,k} \end{bmatrix}$$
$$c_{i,k} = \cos \theta_{i,k}, \quad s_{i,k} = \sin \theta_{i,k}, \quad a_{i,k} = \cos \Delta \theta_{i,k}, \quad b_{i,k} = \sin \Delta \theta_{i,k}$$

## $SO(2)$ Constraints

$$c_{i,k}^2 + s_{i,k}^2 - 1 = 0, \quad a_{i,k}^2 + b_{i,k}^2 - 1 = 0$$

## $SO(3)$ Constraints

$$\mathbf{R}_k = [\mathbf{r}_{1,k} \quad \mathbf{r}_{2,k} \quad \mathbf{r}_{3,k}]$$
$$\|\mathbf{r}_{1,k}\|_2^2 - 1 = \|\mathbf{r}_{2,k}\|_2^2 - 1 = \|\mathbf{r}_{3,k}\|_2^2 - 1 = 0$$
$$\mathbf{r}_{1,k}^\top \mathbf{r}_{2,k} = \mathbf{r}_{2,k}^\top \mathbf{r}_{3,k} = \mathbf{r}_{1,k}^\top \mathbf{r}_{3,k} = 0$$
$$\mathbf{r}_{1,k} \times \mathbf{r}_{2,k} = \mathbf{r}_{3,k}, \quad \mathbf{r}_{2,k} \times \mathbf{r}_{3,k} = \mathbf{r}_{1,k}, \quad \mathbf{r}_{3,k} \times \mathbf{r}_{1,k} = \mathbf{r}_{2,k}$$

The same constraints are imposed on  $\mathbf{F}_k = [\mathbf{f}_{1,k} \quad \mathbf{f}_{2,k} \quad \mathbf{f}_{3,k}]$ .

# 3D-Pendulum Polynomial Optimization Problem

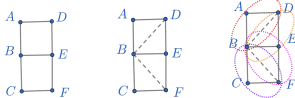
$$\begin{aligned}
 p_P^* = & \min_{\substack{\{\mathbf{R}_k\}_{k=0}^N, \{\mathbf{F}_k\}_{k=0}^{N-1} \\ \{\bar{\mathbf{u}}_{p,k}\}_{k=1}^{N-1}}} w_R \left\| \mathbf{R}_N - \mathbf{R}_N^{\text{des}} \right\|_F^2 + w_F \left\| \mathbf{F}_{N-1} - \mathbf{F}^{\text{des}} \right\|_F^2 \\
 & + \alpha_R \sum_{k=0}^{N-1} \left\| \mathbf{R}_k - \mathbf{R}_N^{\text{des}} \right\|_F^2 + \alpha_F \sum_{k=0}^{N-2} \left\| \mathbf{F}_k - \mathbf{F}^{\text{des}} \right\|_F^2 + \frac{1}{w_u} \sum_{k=1}^{N-1} \|\bar{\mathbf{u}}_{p,k}\|_2^2 \\
 \text{subject to } & \frac{1}{h} \left( \mathbf{F}_{k+1} \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_{k+1}^\top - \mathbf{J}_d \mathbf{F}_k + \mathbf{F}_k^\top \mathbf{J}_d \right) = h (\boldsymbol{\tau}_{g,k+1} + \mathbf{u}_{k+1})^\wedge, & k = 0, \dots, N-2, \\
 & \mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k, & k = 0, \dots, N-1, \\
 & \|\mathbf{r}_{1,k}\|_2^2 - 1 = \|\mathbf{r}_{2,k}\|_2^2 - 1 = \|\mathbf{r}_{3,k}\|_2^2 - 1 = 0, & k = 0, \dots, N, \\
 & \mathbf{r}_{1,k}^\top \mathbf{r}_{2,k} = \mathbf{r}_{2,k}^\top \mathbf{r}_{3,k} = \mathbf{r}_{1,k}^\top \mathbf{r}_{3,k} = 0, & k = 0, \dots, N, \\
 & \mathbf{r}_{1,k} \times \mathbf{r}_{2,k} - \mathbf{r}_{3,k} = \mathbf{r}_{2,k} \times \mathbf{r}_{3,k} - \mathbf{r}_{1,k} = \mathbf{r}_{3,k} \times \mathbf{r}_{1,k} - \mathbf{r}_{2,k} = \mathbf{0}_{3 \times 1}, & k = 0, \dots, N, \\
 & \|\mathbf{f}_{1,k}\|_2^2 - 1 = \|\mathbf{f}_{2,k}\|_2^2 - 1 = \|\mathbf{f}_{3,k}\|_2^2 - 1 = 0, & k = 0, \dots, N-1, \\
 & \mathbf{f}_{1,k}^\top \mathbf{f}_{2,k} = \mathbf{f}_{2,k}^\top \mathbf{f}_{3,k} = \mathbf{f}_{1,k}^\top \mathbf{f}_{3,k} = 0, & k = 0, \dots, N-1, \\
 & \mathbf{f}_{1,k} \times \mathbf{f}_{2,k} - \mathbf{f}_{3,k} = \mathbf{f}_{2,k} \times \mathbf{f}_{3,k} - \mathbf{f}_{1,k} = \mathbf{f}_{3,k} \times \mathbf{f}_{1,k} - \mathbf{f}_{2,k} = \mathbf{0}_{3 \times 1}, & k = 0, \dots, N-1, \\
 & \frac{\text{tr}(\mathbf{F}_k) - 1}{2} \geq \cos(\Delta\theta_{\max}), & k = 0, \dots, N-1, \\
 & 1 - \bar{\mathbf{u}}_{p,k}^\top \bar{\mathbf{u}}_{p,k} \geq 0, & k = 1, \dots, N-1, \\
 & \mathbf{R}_0 = \mathbf{R}^{\text{init}}, \quad \mathbf{F}_0 = \mathbf{F}^{\text{init}}.
 \end{aligned}$$

# Acrobot Polynomial Optimization Problem

$$\begin{aligned}
 p_A^* = \min_{\mathbf{z}_A} & w_R \sum_{i=1}^2 \left\| \mathbf{R}_{i,N} - \mathbf{R}_i^{\text{des}} \right\|_F^2 + w_F \sum_{i=1}^2 \left\| \mathbf{F}_{i,N-1} - \mathbf{F}_i^{\text{des}} \right\|_F^2 \\
 & + \alpha_R \sum_{k=0}^{N-1} \sum_{i=1}^2 \left\| \mathbf{R}_{i,k} - \mathbf{R}_i^{\text{des}} \right\|_F^2 + \alpha_F \sum_{k=0}^{N-2} \sum_{i=1}^2 \left\| \mathbf{F}_{i,k} - \mathbf{F}_i^{\text{des}} \right\|_F^2 + \frac{1}{w_u} \sum_{k=1}^{N-1} \bar{u}_k^2 \\
 \text{subject to} & -M_1 \mathbf{R}_{1,k} \mathbf{A}_{1,k} \boldsymbol{\rho}_{1,0} - h^2 g M_1 \mathbf{e}_2 - h^2 \lambda_{\max} (\bar{\boldsymbol{\lambda}}_{0,k} + \bar{\boldsymbol{\lambda}}_{12,k}) = \mathbf{0}, \quad k = 1, \dots, N-1, \\
 & M_2 [\mathbf{R}_{1,k} \mathbf{A}_{1,k} (\boldsymbol{\rho}_{1,12} - \boldsymbol{\rho}_{1,0}) - \mathbf{R}_{2,k} \mathbf{A}_{2,k} \boldsymbol{\rho}_{2,12}] - h^2 g M_2 \mathbf{e}_2 + h^2 \lambda_{\max} \bar{\boldsymbol{\lambda}}_{12,k} = \mathbf{0}, \quad k = 1, \dots, N-1, \\
 & \hat{\boldsymbol{\Pi}}_{1,k} - \hat{\boldsymbol{\Pi}}_{1,k+1} + h \left[ \boldsymbol{\rho}_{1,0} \times \mathbf{R}_{1,k+1}^\top \lambda_{\max} \bar{\boldsymbol{\lambda}}_{0,k+1} \right. \\
 & \quad \left. + \boldsymbol{\rho}_{1,12} \times \mathbf{R}_{1,k+1}^\top \lambda_{\max} \bar{\boldsymbol{\lambda}}_{12,k+1} - u_{\max} \bar{u}_{k+1} \right]^\wedge = \mathbf{0}, \quad k = 0, \dots, N-2, \\
 & \hat{\boldsymbol{\Pi}}_{2,k} - \hat{\boldsymbol{\Pi}}_{2,k+1} + h \left[ u_{\max} \bar{u}_{k+1} - \boldsymbol{\rho}_{2,12} \times \mathbf{R}_{2,k+1}^\top \lambda_{\max} \bar{\boldsymbol{\lambda}}_{12,k+1} \right]^\wedge = \mathbf{0}, \quad k = 0, \dots, N-2, \\
 & \mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}, \quad i = 1, 2, \quad k = 0, \dots, N-1, \\
 & c_{i,k}^2 + s_{i,k}^2 - 1 = 0, \quad i = 1, 2, \quad k = 0, \dots, N, \\
 & a_{i,k}^2 + b_{i,k}^2 - 1 = 0, \quad i = 1, 2, \quad k = 0, \dots, N-1, \\
 & a_{i,k} - \cos(\Delta\theta_{i,\max}) \geq 0, \quad i = 1, 2, \quad k = 1, \dots, N-1, \\
 & 1 - \bar{u}_k^2 \geq 0, \quad k = 1, \dots, N-1, \\
 & 1 - \left( \bar{\lambda}_{0,k}^{(j)} \right)^2 \geq 0, \quad 1 - \left( \bar{\lambda}_{12,k}^{(j)} \right)^2 \geq 0, \quad j = 1, 2, \quad k = 1, \dots, N-1, \\
 & \mathbf{F}_{i,0} = \mathbf{F}_i^{\text{init}}, \quad \mathbf{R}_{i,1} = \mathbf{R}_i^{\text{init}}, \quad i = 1, 2, \\
 & \mathbf{A}_{i,k} = \mathbf{F}_{i,k} - 2\mathbf{I}_2 + \mathbf{F}_{i,k-1}^\top.
 \end{aligned}$$

# Sparse Clique Construction

## 3D Pendulum: Automatic Minimum-Degree Chordal Extension

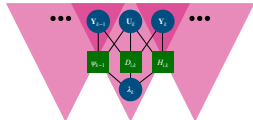


Original graph  $\rightarrow$  MD extension  $\rightarrow$  maximal cliques

**SPOT automatically constructs the correlative sparsity graph, adds fill edges using MD, and extracts its maximal cliques.**

$$\begin{array}{ccc} |I_{\max}^P| = 24, & M_{2,\max}^P \in \mathbb{S}^{325} \\ \underbrace{91,378 \times 91,378}_{\text{dense, } N=20} & \longrightarrow & \underbrace{325 \times 325}_{\text{largest MD block}} \end{array}$$

## Acrobot: Complete Chain-Like Cliques



$$z_A(I_k^A) = [R_{1,k} \ R_{2,k} \ R_{1,k+1} \ R_{2,k+1} \ F_{1,k-1} \ F_{2,k-1} \ F_{1,k} \ F_{2,k} \ \bar{\lambda}_{0,k} \ \bar{\lambda}_{12,k} \ \bar{u}_k]^\top$$

One clique contains both links and two neighboring time steps.

$$\begin{array}{ccc} \underbrace{351 \times 351} & & \underbrace{253 \times 253} \\ |I_1^A| = 25 & & |I_k^A| = 21, \ k \geq 2 \end{array}$$

## Acrobot: Multibody Cliques

Split each interior stage into a coupled dynamics clique and a kinematics clique:

$$z_A(I_{\text{dyn},k}^A) = [R_{1,k} \ F_{1,k-1} \ F_{1,k} \ R_{2,k} \ F_{2,k-1} \ F_{2,k} \ \bar{\lambda}_{0,k} \ \bar{\lambda}_{12,k} \ \bar{u}_k]^\top \in \mathbb{R}^{17}$$

$$z_A(I_{\text{kin},k}^A) = [R_{1,k} \ R_{2,k} \ R_{1,k+1} \ R_{2,k+1} \ F_{1,k} \ F_{2,k}]^\top \in \mathbb{R}^{12}$$

The first two complete cliques are retained around the first control.

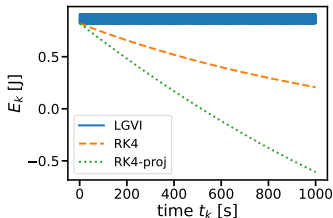
$$\begin{array}{ccc} \underbrace{351, 253} & \longrightarrow & \underbrace{171, 91} \\ \text{first two blocks} & & \text{interior blocks} \end{array}$$

# Numerical Simulation of 3D-Pendulum

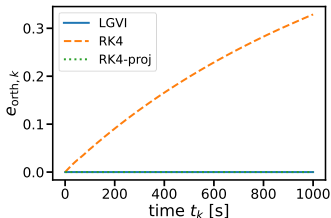
## Unforced:

- $u_k = 0$
- $h = 0.05$  s
- $t_k \in [0, 1000]$  s

**LGVI preserves energy**  
over long time horizons



(a) Total energy  $E_k$ .

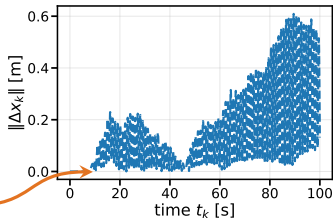


(b) Orthogonality error  $e_{\text{orth},k}$ .

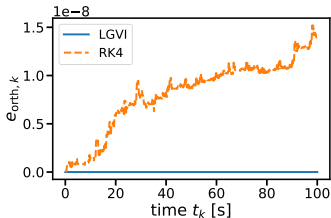
## Controlled:

- $u_k = R_k^\top e_3 \times u_{p,k}$
- $h = 0.001$  s
- $t_k \in [0, 100]$  s

**RK4-LGVI**  
 $\Delta x_k$  deviation after short time



(c) Position error norm  $\Delta x_k$ , with  $x_k = R_k \rho_C$ .



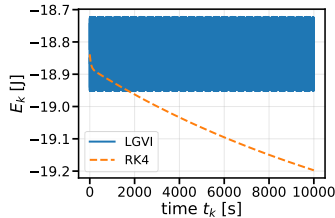
(d) Orthogonality error  $e_{\text{orth},k}$ .

# Numerical Simulation of Acrobot

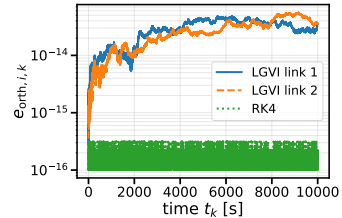
## Unforced:

- $u_k = 0$
- $h = 0.01$  s
- $t_k \in [0, 10000]$  s

**LGVI preserves energy**  
over long time horizons



(a) Total energy  $E_k$ .

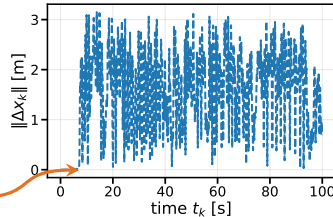


(b) Orthogonality error  $e_{\text{orth},i,k}$ ,  $i \in \{1, 2\}$ .

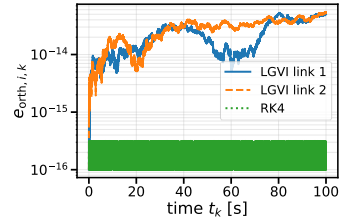
## Controlled:

- $u_k = 0.15 \sin(0.7t_k)$
- $h = 0.001$  s
- $t_k \in [0, 100]$  s

**RK4-LGVI**  
 $\|\Delta \mathbf{X}_k\|_2$  deviation  
after  
short time



(c) Position error norm  $\|\Delta \mathbf{X}_k\|_2$ .



(d) Orthogonality error  $e_{\text{orth},i,k}$ ,  $i \in \{1, 2\}$ .

# Acrobot Optimization Results

Open-loop SDP relaxation,  $\kappa = 2$ ,  $N = 60$

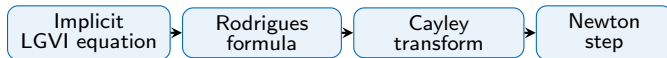
| Metric                                     | Chain sparsity        |        | Multibody sparsity   |               |
|--------------------------------------------|-----------------------|--------|----------------------|---------------|
|                                            | 45°                   | 180°   | 45°                  | 180°          |
| Gap $\epsilon$                             | $4.8 \times 10^{-10}$ | 0.5233 | $9.2 \times 10^{-9}$ | 0.5668        |
| Rank $\delta_{\max}$                       | $1.1 \times 10^{-8}$  | 0.5596 | $2.4 \times 10^{-6}$ | <b>0.6603</b> |
| First-clique rank $\delta_1$               | $2.8 \times 10^{-10}$ | 0.2066 | $2.4 \times 10^{-8}$ | <b>0.2172</b> |
| Manifold error $e_{\text{orth}}^{A, \max}$ | $1.9 \times 10^{-8}$  | 1.4053 | $3.4 \times 10^{-6}$ | 1.4561        |
| Runtime [min]                              | 59.7                  | 90.4   | <b>17.6</b>          | <b>31.2</b>   |

## Key Findings

- I. Chain  $\rightarrow$  multibody sparsity:  $\approx 3\times$  **faster**
- II.  $\delta_1 < \delta_{\max}$ : the first clique is **tighter** than the worst trajectory clique

# Computational Implementation

**Goal:** compute the relative attitude update  $\mathbf{F}_k \in SO(3)$  from the implicit discrete equations.



## Rodrigues formula:

$$\mathbf{F} = \exp(\hat{\mathbf{f}}) = \mathbf{I} + \frac{\sin \|\mathbf{f}\|}{\|\mathbf{f}\|} \hat{\mathbf{f}} + \frac{1 - \cos \|\mathbf{f}\|}{\|\mathbf{f}\|^2} \hat{\mathbf{f}}^2$$

- Converts  $\mathbf{F}_k \in SO(3)$  into  $\mathbf{f}_k \in \mathbb{R}^3$ .
- Gives implicit equation:

$$\mathbf{0} = \mathbf{A}(\mathbf{f}_k), \quad \mathbf{A} : \mathbb{R}^3 \rightarrow \mathbb{R}^3.$$

## Cayley transformation:

$$\begin{aligned} \mathbf{F} &= \text{Cay}(\mathbf{f}) = (\mathbf{I} + \hat{\mathbf{f}})(\mathbf{I} - \hat{\mathbf{f}})^{-1} \\ &= \frac{1}{1 + \mathbf{f}^\top \mathbf{f}} \left( (1 - \mathbf{f}^\top \mathbf{f})\mathbf{I} + 2\hat{\mathbf{f}} + 2\mathbf{f}\mathbf{f}^\top \right) \end{aligned}$$

- Efficient local parametrization for small rotation updates.
- Avoids  $\sin(\cdot)$ ,  $\cos(\cdot)$ .
- Valid locally:  $\|\mathbf{f}\| = \tan(\theta/2)$ ,  $\theta < \pi$ .

### Implicit solve in $\mathbb{R}^3$

$$\mathbf{f}^{(i+1)} = \mathbf{f}^{(i)} - \nabla \mathbf{A}(\mathbf{f}^{(i)})^{-1} \mathbf{A}(\mathbf{f}^{(i)}), \quad \left\| \mathbf{f}^{(i+1)} - \mathbf{f}^{(i)} \right\| < \varepsilon$$

$$\text{After convergence: } \mathbf{F}_k = \text{Cay}(\mathbf{f}_k), \quad \mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k.$$